

Experimental Design

1.1 Planning and Conducting Experiments

Surveys tell you what people think or report. Experiments tell you what actually causes what. The distinction determines what conclusions you are entitled to draw when the data comes in.

1.1.1 Experiments vs. Observational Studies

The defining question is whether the researcher assigns the conditions being studied or simply observes them as they exist.

In an *observational study*, the researcher collects data on subjects without intervening. The variables of interest already exist in the subjects; the researcher records what is there. A sociologist who tracks the grades and sleep hours of university students over a semester, without changing anyone's sleep schedule, is conducting an observational study.

In an *experiment*, the researcher imposes conditions on subjects and then measures the effect or change. The researcher creates the differences rather than observing them.

Why does the distinction matter? Only a properly designed experiment can support a cause-and-effect conclusion. An observational study can show that two variables are associated, but a third variable you did not account for could explain the relationship entirely. This uninvited third variable is called a *confounding variable*: one whose effect on the response cannot be separated from the effect of the variable you are actually studying.

In the observational sleep study, students who sleep more might also exercise more, eat better, or have lower course loads. Any of these could be responsible for better grades. There is no way to figure out which effects correlate to which behaviours.

There are situations where an experiment is unethical or impossible: you cannot randomly assign people to smoke for twenty years, or randomly select cities to experience floods. In these cases, observational data is all we have, and conclusions must be drawn with caution.

Exercise 1 Experiment or Observational Study?

For each study, identify whether it is an experiment or an observational study. Then state whether a cause-and-effect conclusion is justified and why.

Working Space

Study	Type	Causal conclusion?
Researchers randomly assign 180 seedlings to three groups and water each group with a different fertilizer solution. After eight weeks they measure plant height.		
A researcher records the screen time and attention span of 500 children aged 8 to 12, then looks for a relationship between the two.		
A company randomly selects half its customer service team to complete a new communication training program, then compares customer satisfaction scores between the two groups.		
A historian finds that cities with more libraries per capita had higher literacy rates in the early 20th century.		

Answer on Page 7

1.1.2 Vocabulary for Experiments

Experiments have their own vocabulary. These terms build on each other, so it helps to learn them in order.

Explanatory variable Also called the independent variable. The variable whose effect on the response is being studied. In a study of whether LoFi music affects study focus, the type of content being studied is the explanatory variable.

Response variable Also called the dependent variable. The outcome being measured. In a study of whether background music affects study typing speed, the typing speed is the response variable.

Experimental unit The smallest unit to which a treatment is applied. In a study of plants, each plant is an experimental unit. In a study of people, each person is an experimental unit.

Factor A variable whose effect on the response is of interest. Factors can be qualitative (type of music: classical, jazz, Afrobeat) or quantitative (temperature in degrees Kelvin).

Levels The specific values of a factor used in the experiment. If the factor is type of music with three options, there are three levels.

Treatment A specific combination of factor levels applied to experimental units. With one factor and three levels, there are three treatments. With two factors at two levels each, there are $2 \times 2 = 4$ treatments.

Control group A group that receives no treatment or a standard baseline condition. It provides a reference point for comparison with the treatment groups.

Placebo A dummy treatment that resembles the real treatment but contains no active component. Used to isolate the genuine effect of the treatment from the *placebo effect*: the tendency for people to show change simply because they believe they are receiving treatment.

1.1.3 The Three Principles of Experimental Design

A well-designed experiment rests on three principles.

Control means keeping everything the same for all groups except the treatment. It is not just having a control group. Control helps make sure the treatment is the only real difference.

Randomization means using chance to place experimental units into groups. It helps keep the groups fair and spreads unknown factors across all groups.

A *lurking variable* is something that can affect the response but is not part of the experiment.

Replication means including enough experimental units in each group so that chance variation among individuals does not swamp the treatment effect. If a new training method is tested on only two employees per group, a single unusually high or low performer could dominate the group's result. With 30 employees per group, individual extremes average out.

Exercise 2 Applying the Three Principles

A horticulture researcher wants to test whether exposing seedlings to classical music for two hours per day increases their growth rate compared to no music exposure. She has 60 seedlings of the same species, grown from the same seed batch, available for the study.

Working Space

1. Describe how randomization should be applied.
2. What does control mean in this context? List at least three variables that should be held constant.
3. The researcher initially plans to use 5 seedlings per group. A colleague recommends 30 per group. Which principle does this change address, and why?
4. Should there be a control group? What would it look like?

Answer on Page 7

1.1.4 Blinding

When subjects know which treatment they are receiving, their behavior or reported experience may change in ways that have nothing to do with the treatment itself. When the person measuring outcomes knows which group a subject belongs to, their assessments may be subtly influenced. Both of these are sources of bias that blinding is designed to prevent.

In a *single-blind experiment*, either the subjects do not know which treatment they received, or the person measuring outcomes does not know, but not both. In a *double-blind experiment*, neither the subjects nor the evaluators know the treatment assignments. Double-blind experiments are preferred when feasible because they eliminate bias from

both sources simultaneously.

Double-blinding is not always possible. If subjects can easily identify their treatment condition from its appearance or feel, blinding them is impossible. A study comparing two different exercise regimens cannot prevent participants from knowing which regimen they are following. In such cases, the best available design is to blind the evaluators, even if the subjects cannot be blinded.

This is a draft chapter from the Kontinua Project. Please see our website (<https://kontinua.org/>) for more details.

Answers to Exercises

Answer to Exercise 1 (on page 2)

- **Seedlings:** Experiment. Fertilizer was randomly assigned. Causal conclusion is justified.
- **Screen time and attention:** Observational study. No assignment was made. Causal conclusion is not justified; many confounding variables (parenting style, diet, sleep) could explain the relationship.
- **Customer service training:** Experiment. The training was randomly assigned. Causal conclusion is justified.
- **Libraries and literacy:** Observational study. No assignment was made. Causal conclusion is not justified; wealthier cities may have had both more libraries and higher literacy for unrelated reasons.

Answer to Exercise 2 (on page 4)

1. Number the 60 seedlings from 1 to 60. Use a random number generator to select 30 for the music group. The remaining 30 form the no-music group.
2. Control means growing all seedlings under identical conditions except for the music. Variables to hold constant include: soil type and quantity, pot size, amount and frequency of watering, light exposure, temperature, and humidity.
3. Replication. With only 5 seedlings per group, a single unusually fast or slow grower could distort the group average. With 30 per group, individual variation averages out and a real effect of the music is more detectable.
4. Yes. The no-music group serves as the control group, establishing the baseline growth rate without the treatment. Without it, there is nothing to compare the music group's growth against.



INDEX

confounding variable, 1

Control, 3

control

 in experiments, 3

Control group, 3

control group, 3

double-blind experiment, 4

experiment, 1

Experimental unit, 3

experimental unit, 3

Explanatory variable, 2

explanatory variable, 2

Factor, 3

factor, 3

Levels, 3

levels, 3

lurking variable, 3

observational study, 1

Placebo, 3

placebo, 3

placebo effect, 3

Randomization, 3

randomization, 3

Replication, 3

replication, 3

Response variable, 2

response variable, 2

single-blind experiment, 4

Treatment, 3

treatment, 3